# 1 I

## 1.1 (1)

At $T = 0$, the Fermi–Dirac distribution reduces to a step function:

$$f(\epsilon) = \theta(\epsilon_F - \epsilon).$$

For $B = 0$, the single-electron Hamiltonian is:

$$h_0(\vec{p}) = \frac{|\vec{p}|^2}{2m}.$$

The Fermi momentum is therefore $p_F = \sqrt{2m\epsilon_F}$.

Thus, the number of electrons with energy below $\epsilon_F$ for a given spin is:

$$\begin{aligned} N_0(\epsilon_F) &= \int dn_x\, dn_y \\ &= \left(\frac{L}{2\pi\hbar}\right)^2 \int_{p_x^2+p_y^2\le p_F^2} dp_x\, dp_y \\ &= \left(\frac{L}{2\pi\hbar}\right)^2 2\pi \int_0^{p_F} p\, dp \\ &= \frac{Sm}{2\pi\hbar^2}\epsilon_F. \end{aligned}$$

## 1.2 (2)

The magnetic moment of an electron with spin $\sigma$ is $\mu_B\sigma$. Thus, the total spin magnetization is:

$$M = \mu_B(N_+ - N_-).$$

At $T = 0$, the numbers of electrons with spin up and spin down are:

$$\begin{aligned} N_+ &= N_0(\epsilon_F + \mu_B B), \\ N_- &= N_0(\epsilon_F - \mu_B B). \end{aligned}$$

Therefore:

$$\begin{aligned} M &= \mu_B \left[N_0(\epsilon_F + \mu_B B) - N_0(\epsilon_F - \mu_B B)\right] \\ &= \boxed{\frac{Sm\mu_B^2}{\pi\hbar^2} B}. \end{aligned}$$

# 2 II

## 2.1 (3)

The single-electron partition function is:

$$z = \frac{1}{(2\pi\hbar)^2} \int_S dx\, dy \int_{-\infty}^{\infty} dp_x \int_{-\infty}^{\infty} dp_y\, e^{-\frac{\beta}{2m}[p_x^2 + (p_y + qBx)^2]}.$$

Performing the momentum integrations first, we obtain:

$$\begin{aligned} z &= \frac{1}{(2\pi\hbar)^2} \int_S dx\, dy \sqrt{\frac{2\pi m}{\beta}} \sqrt{\frac{2\pi m}{\beta}} \\ &= \boxed{\frac{Sm}{2\pi\hbar^2\beta}}. \end{aligned}$$

### 2.2 (4)

Since the single-electron partition function $z$ is independent of $B$, the total partition function $Z = z^N/N!$ is also independent of $B$.

Therefore, the Helmholtz free energy $F = -k_B T \ln Z$ is independent of $B$, and:

$$\boxed{M = -\frac{\partial F}{\partial B} = 0}.$$

## 3 III

### 3.1 (5)

For an arbitrary single-electron state with energy $\epsilon_l$, the occupation number can only be $n_l = 0$ or $n_l = 1$.

Thus, the contribution of this state to the grand partition function is:

$$\xi_l = \sum_{n_l=0}^{1} e^{-\beta(\epsilon_l - \mu)n_l} = 1 + e^{-\beta(\epsilon_l - \mu)}.$$

Since the occupation numbers of different states are independent, we can multiply their contributions to obtain the grand partition function:

$$\boxed{\Xi = \prod_{l=0}^{\Lambda} \left(1 + e^{-\beta(\epsilon_l - \mu)}\right)}.$$

### 3.2 (6)

From the grand potential $J = -k_B T \ln \Xi$, we obtain:

$$\begin{aligned} N &= -\frac{\partial J}{\partial \mu} \\ &= \boxed{\sum_{l=0}^{\Lambda} \frac{1}{1 + e^{\beta(\epsilon_l - \mu)}}}. \end{aligned}$$

## 4 IV

### 4.1 (7)

The single-electron energy levels and their degeneracy are:

$$\epsilon_l = 2\mu_B B \left(l + \frac{1}{2}\right), \qquad D = \frac{qBS}{2\pi\hbar}.$$

Therefore, the grand potential is:

$$\begin{aligned} J &= -k_B T \ln \Xi \\ &= -k_B T D \sum_{l=0}^{\infty} \ln \left[1 + e^{-\beta(2\mu_B B(l+1/2) - \mu)}\right]. \end{aligned}$$

Comparing this with $J = \sum_{l=0}^{\infty} \phi(l + 1/2)$, we obtain:

$$\boxed{\phi(x) = -\frac{k_B T q B S}{2\pi\hbar} \ln\left[1 + e^{-\beta(2\mu_B B x - \mu)}\right]}.$$

## 4.2 (8)

The first term in the Euler–Maclaurin formula is:

$$J_0 = \int_0^{\infty} \phi(x)\, dx.$$

Substituting the result from (7):

$$J_0 = -\frac{k_B T q B S}{2\pi\hbar} \int_0^{\infty} \ln\left[1 + e^{-\beta(2\mu_B B x - \mu)}\right] dx.$$

Introducing a new variable $u = 2\mu_B B x$, we have:

$$dx = \frac{du}{2\mu_B B}.$$

Thus:

$$J_0 = -\frac{k_B T q S}{4\pi\hbar\mu_B} \int_0^{\infty} \ln\left[1 + e^{-\beta(u - \mu)}\right] du.$$

Since $\mu_B = \hbar q/(2m)$, we obtain:

$$\boxed{J_0 = -\frac{S m k_B T}{2\pi\hbar^2} \int_0^{\infty} \ln\left[1 + e^{-\beta(u - \mu)}\right] du}.$$

Therefore, $J_0$ is independent of $B$.

## 4.3 (9)

Differentiating $\phi(x)$ with respect to $x$, we obtain:

$$\phi'(x) = \frac{q S \mu_B B^2}{\pi\hbar} \frac{1}{1 + e^{\beta(2\mu_B B x - \mu)}}.$$

Thus:

$$\phi'(\infty) = 0,$$
$$\phi'(0) = \frac{q S \mu_B B^2}{\pi\hbar} \frac{1}{1 + e^{-\beta\mu}}.$$

Using the Euler–Maclaurin formula, the grand potential up to the second order in $B$ is:

$$\begin{aligned} J &\simeq J_0 - \frac{1}{24}\left[\phi'(\infty) - \phi'(0)\right] \\ &= J_0 + \frac{q S \mu_B B^2}{24\pi\hbar} \frac{1}{1 + e^{-\beta\mu}}. \end{aligned}$$

Since $J_0$ is independent of $B$, we finally obtain:

$$\boxed{M = -\left(\frac{\partial J}{\partial B}\right)_{T,\mu} = -\frac{q S \mu_B B}{12\pi\hbar} \frac{1}{1 + e^{-\beta\mu}}}.$$